

Robert, 55 · sudden quadriplegia · cannot speak or swallow · awake and aware · communicates with vertical eye movements only · MRI: **infarct of the ventral pons** · **locked-in syndrome from basilar artery occlusion**

BIO 004 · WEEK 7 · CASE DEEP DIVE

The Brainstem

A patient awake but completely paralyzed except for vertical eye movements. One stroke. One small structure with enormous consequences.

PATIENT CASE

Robert is 55. He collapsed at work and was rushed to the ED. On arrival he is **quadriplegic, cannot speak, cannot swallow**, yet his eyes are open and tracking. He can move them **vertically only**. When asked to blink twice for yes, he does.

MRI brain shows an **infarct of the ventral pons** from **basilar artery occlusion**. The diagnosis is **locked-in syndrome**. Robert's consciousness, cognition, and ability to see and hear are intact. He has lost all voluntary motor control except vertical gaze and blinking. The team begins working with him on an eye-tracking communication system.

The brainstem is the smallest division of the brain but the most densely packed with critical structures. Robert's case shows what happens when one small region of it fails.

GUIDING QUESTION

What are the three regions of the brainstem, what passes through them, and how can one ventral pontine stroke leave the patient awake but completely motorless?

1 THREE REGIONS

From top to bottom

Midbrain

Connects diencephalon to pons. Contains cerebral peduncles (descending motor fibers), tectum (superior + inferior colliculi for visual and auditory reflexes), substantia nigra. CN III and IV nuclei.

Pons

Bulging middle section. "Bridge" between cerebellum and cerebrum. CN V, VI, VII, VIII nuclei. Pneumotaxic and apneustic respiratory centers. *Robert's lesion.*

Medulla oblongata

Lowest. Continuous with spinal cord. Pyramids (corticospinal tract decussation), olives (motor relays), CN IX-XII nuclei. Cardiovascular and respiratory centers.

2 THREE THINGS PASSING THROUGH

Ascending, descending, cranial

The brainstem is a small region but every structure that needs to go between brain and body passes through it. Three categories:

Descending motor tracts

Corticospinal (voluntary movement), corticobulbar (to cranial nerve nuclei), corticopontine. *Robert's lesion destroys these.*

Ascending sensory tracts

Medial lemniscus (fine touch, proprioception), spinothalamic (pain, temperature). These run more dorsally in the brainstem and may be spared in ventral lesions.

Cranial nerve nuclei

Ten of the 12 cranial nerve nuclei sit in the brainstem (CN III-XII). Lesions cause specific cranial nerve palsies that help localize the level.

Which CN nuclei live where

REGION	CRANIAL NERVE NUCLEI
Midbrain	III (oculomotor), IV (trochlear)
Pons	V (trigeminal motor + main sensory), VI (abducens), VII (facial), VIII (vestibulocochlear)
Medulla	IX (glossopharyngeal), X (vagus), XI (accessory — partial), XII (hypoglossal)

Diffuse network controlling consciousness

Throughout the brainstem, scattered neurons form the **reticular formation**, a diffuse network controlling consciousness, arousal, and autonomic functions. The *ascending reticular activating system (ARAS)* projects to the cerebral cortex via the thalamus and maintains wakefulness. Brainstem strokes that disrupt ARAS cause coma; lesions sparing ARAS (like Robert's ventral pons) leave consciousness intact.

Cardiac, respiratory, GI

The medulla contains nuclei that control **cardiac output** (heart rate, contractility, vasomotor tone), **respiration** (basic breathing rhythm via the medullary respiratory center; modulated by pons), **swallowing**, **vomiting**, and **coughing**. Damage to these regions is rapidly fatal. This is why brain death is defined in part by loss of brainstem function.

The most striking dissociation in neurology

Why Robert is locked in. The ventral pons contains the descending corticospinal and corticobulbar tracts running from cortex to spinal cord and to cranial nerve nuclei. A ventral pontine stroke destroys these fibers — Robert's brain still gives motor commands, but they can't get out. Most cranial nerves below the lesion (V, VII, IX, X, XII) are paralyzed (no speech, no swallow, no facial movement). What's SPARED: the ARAS (consciousness intact), the dorsal sensory tracts (he can feel), and CN III/IV/VI in the midbrain (vertical eye movement preserved because the midbrain is above the lesion; horizontal gaze involves a pontine center that's damaged).



DRAW ZONE 1 -- Sketch the brainstem from the front showing midbrain, pons, medulla. Label the cranial nerves emerging from each region.

CN I and II don't come from the brainstem (they're forebrain). The other 10 do.



DRAW ZONE 2 -- Diagram a cross-section of the pons showing the ventral (basis pontis) and dorsal (tegmentum) regions. Mark where Robert's stroke is and what passes through.

Ventral = corticospinal/corticobulbar tracts. Dorsal = sensory tracts + reticular formation + CN nuclei. His ventral stroke spares everything dorsal — which is why he's conscious.

RETURNING TO ROBERT

The most precise dissociation in neuroanatomy

Robert's case is anatomy weaponized: a stroke in the smallest possible region produces the most striking deficit in medicine.

"Quadriplegia from a pontine stroke"

Corticospinal tracts running through the ventral pons are destroyed. All voluntary motor signals to limbs and trunk blocked.

"Loss of speech, swallow, facial movement"

Corticobulbar tracts to CN V, VII, IX, X, XII nuclei are blocked. The muscles themselves are healthy but receive no command.

"Preserved consciousness"

Ascending reticular activating system runs through the dorsal pons (tegmentum), spared by Robert's ventral lesion. He is fully aware.

"Preserved sensation"

Medial lemniscus and spinothalamic tracts run more dorsally. Spared by ventral pontine stroke. He feels everything.

"Preserved vertical eye movement, lost horizontal"

Vertical gaze center is in the midbrain (above the lesion). Horizontal gaze involves the paramedian pontine reticular formation (in the affected pons). Result: he can look up/down but not side to side.

"Communication"

Eye-tracking devices, blink patterns, or simply yes/no via blinks let locked-in patients communicate effectively. Patients have written entire memoirs this way.

Pons, ventral tracts, dorsal sparing, ARAS. Robert's tragedy is also the cleanest demonstration that a few millimeters of brainstem dictate which functions live or die.

